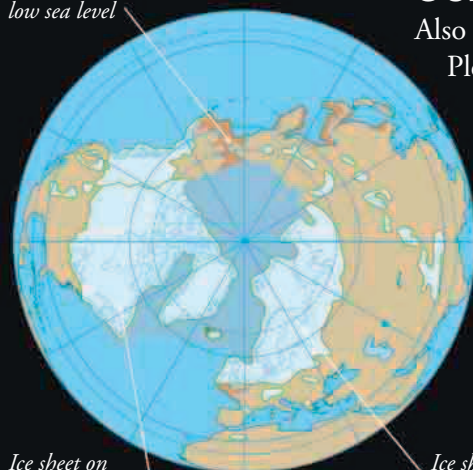


Ice ages

Earth has experienced several ice ages, when the climate was icy cold over large areas. Each ice age has had warm and cold periods. Right now, we are in a warm period of an ice age. During the last cold period, glaciers extended across much of northern Europe and North America. Today, there are still permanent ice sheets in Greenland and Antarctica.

Land created by
low sea level



Ice sheet on
North America

OUR ICE AGE

Also known as the Quaternary or Pleistocene glaciation, our ice age started 2.6 million years ago. The ice has ebbed and flowed several times since. It was at its peak just 18,000 years ago, as shown here, when Arctic ice sheets covered much of North America and northern Eurasia.

Ice sheet on
northern Eurasia

MAMMOTH STEPPE

Although ice sheets covered vast areas during the last big freeze, 18,000 years ago, there were also large areas of snowy tundra and open grassland in northern Europe and Asia. These grasslands, known as mammoth steppe, supported herds of animals such as mammoths. The herds were hunted by powerful predators, including saber-toothed cats, and also humans, who painted images of their prey on cave walls.

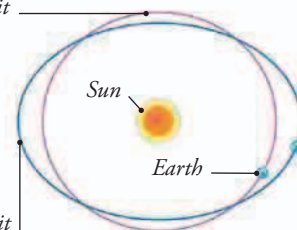


ORBITAL CYCLES

Ice ages are caused by continental drift, which moves land into colder parts of the globe, and by variations in Earth's orbit around the Sun. There are three orbital cycles. The first, a 100,000-year cycle, causes Earth's orbit to change from circular to elliptical. The second, a 42,000-year cycle, affects the tilt of Earth's axis. The third, a 25,800-year cycle, causes Earth to wobble on its axis. Sometimes these cycles combine and chill the planet enough to make the ice sheets start growing.

Circular
Earth orbit

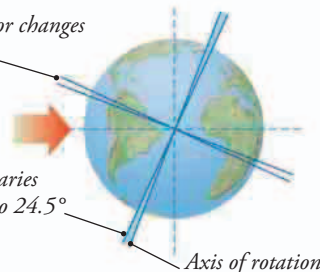
Elliptical
Earth orbit



Orbit shape: 100,000-year cycle

Tilt of equator changes
during cycle

Tilt of axis varies
from 21.6° to 24.5°



Tilt: 42,000-year cycle

Solar
energy

Axis of
rotation



Wobble: 25,800-year cycle

◀ CAVE CREATURES

This image of a horse was painted on a cave wall in Lascaux, France, around 13,000–15,000 BCE during the last cold period of the ice age.